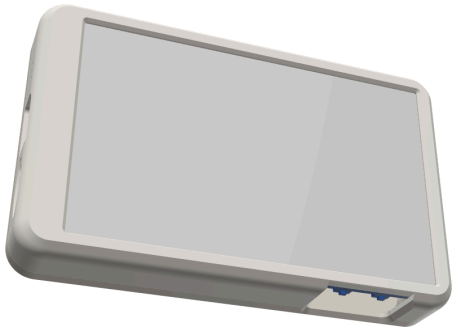
# Case for the ESP32-S3-ePaper-3.97: assembly guide

What to buy, how to print it and what goes where



A two-part 3D-printed case (104.9 × 65.4 × 16.7 mm) for the Waveshare ESP32-S3-ePaper-3.97 board. The battery and the speaker sit side by side inside it, each in a bay made for **your own** battery and speaker, measured from your photos. The board is **screwed to the back cover** through its four brass corner standoffs, with four screws from the back, and the **front lid**, a frame round the display, **snaps on** with four clips. Nothing covers the display, and the lid never presses the board or the glass. This version is printed in PLA.

## What you need to buy

| Item | Qty | Notes |
|---|---|---|
| **Waveshare ESP32-S3-ePaper-3.97** | 1 | The board itself (SKU 33552). |
| **Waveshare 3.7 V 1500 mAh LiPo battery, model 454261** | 1 | The bay is made for yours: measured 61.1 × 40.7 × 4.3 mm, bay 62.5 × 42.5 × 5 mm, so it goes in easily but can only shift about 0.9 mm. |
| **The speaker that came with the board** | 1 | The pocket is made for yours: measured 14.2 × 10.2 × 4.5 mm, cable about 10–11 cm. It has about 0.5 mm of room on every side. |
| **M2.5 × 8 mm pan-head screws (ISO 7045 or DIN 7985)** | 4 | **Exactly 8 mm** under the head. Pan head 5.0 mm across and no more than 2.0 mm tall, flat underneath. **Not cheese or button heads**: they are smaller, and the floor of each recess has a slot across it (so the hole prints without supports), so a smaller head squashes the plastic. They go through the back cover's corners into the board's brass standoffs. |
| **Only if your standoffs are M2: M2 × 8 mm screws and M2 washers** | 4 + 4 | Same kind of screw. Washers 5 mm across, 0.3 mm thick (ISO 7089 / DIN 125), one under each head. Step 4 tells you how to find out. |
| **Thin double-sided tape (0.1–0.2 mm)** | a few cm | One piece under the battery, and a frame round the speaker's membrane (step 2). |

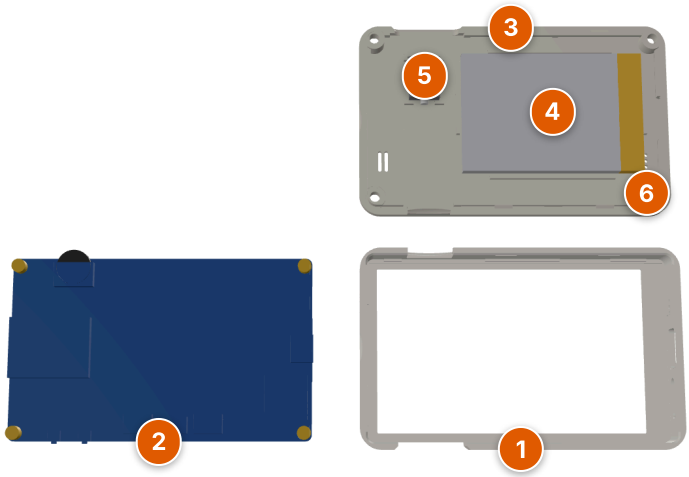
| | | |
|---|---|---|
| **PLA filament** | about 27 g | For both printed parts (PrusaSlicer's estimate). PETG or PCTG also work with no change, and take heat better. |

### Tools

- A small screwdriver that fits your screws (usually Phillips PH1 for M2.5, PH0 for M2)
- A fingernail or a guitar pick, to open the lid
- Scissors or a craft knife for the tape

## The parts

Everything that goes into the case, laid out with the inside of each part facing you:




1 Front lid (printed): the frame round the display. The board goes in here, display side down.

2 The board, seen from its back (components side), with its 4 brass standoffs.

3 Back cover (printed).

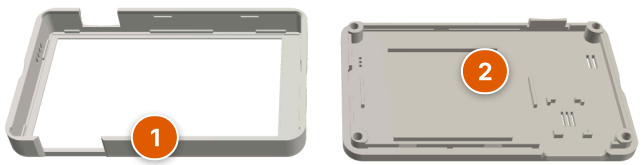
4 Battery in its bay (guides on all four sides).

5 Speaker in its pocket, over the three grille slots.

6 Corner posts: they stand on the board's standoffs, and the screws go through them.

## Printing

- Material: **PLA**. It softens at about 55 °C: keep the case out of direct sun and hot cars. It is also more brittle than PETG, so don't over-tighten the screws. PETG or PCTG work too, with no change.
- Layer height: **0.2 mm or finer** (0.1–0.2 mm). Don't use 0.3 mm.
- 3 perimeters (walls), 15–20 % infill.
- **No supports needed.** Keep the part cooling fan on: each clip beam prints as a 28 mm bridge over its slot.
- Front lid: print it **display side down**. Back cover: the file is already turned over; print it **outside face down**.




1 Front lid: display side down on the bed.

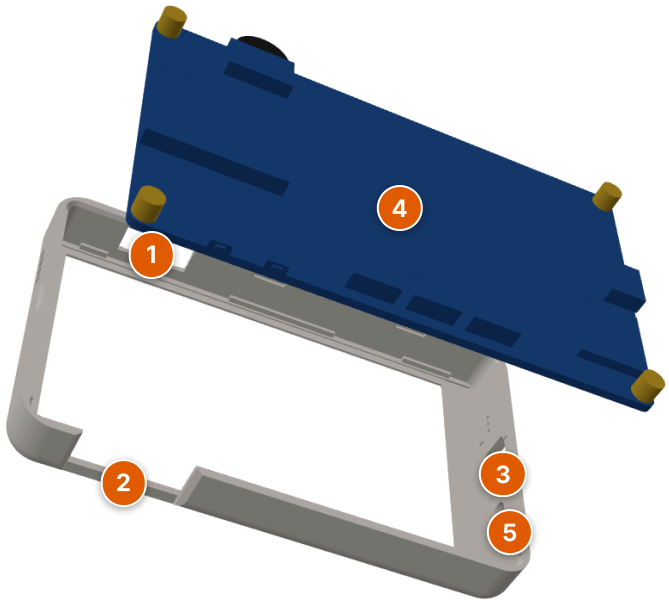
2 Back cover: outside face down on the bed.

**Print the back cover first** (about 15 cm³). When it's done, press the middle of each beam inward: it should move, and its slot should be open along its whole length (if a bridge sagged and stuck somewhere, run a craft knife's blade along the slot). Then lay it on the back of the board: through each corner hole you should see the middle of a standoff's threaded hole, and each screw should start a few turns by hand. Then print the lid. If you use ASA or ABS, turn on your slicer's XY shrinkage compensation for that filament: those plastics shrink enough to make the board too tight.

## Assembly

### 1 Put the board into the front lid

If the display still has its clear protective film (a red pull tab at one corner of the display's left end, where its flex cable is), **peel it off first**: once the board is in the lid, the tab is hidden under the frame and can get caught between the board's edge and the lid's wall. Lay the front lid display side down. Put the board in **display side down**. The dial drops into the notch on the top edge, the PWR/BOOT buttons sit in the opening in the bottom edge, and the USB-C and microSD ports line up with their openings in the end. The board rests on small pads round its edge, which keep the glass off the lid. The picture is seen from the back.

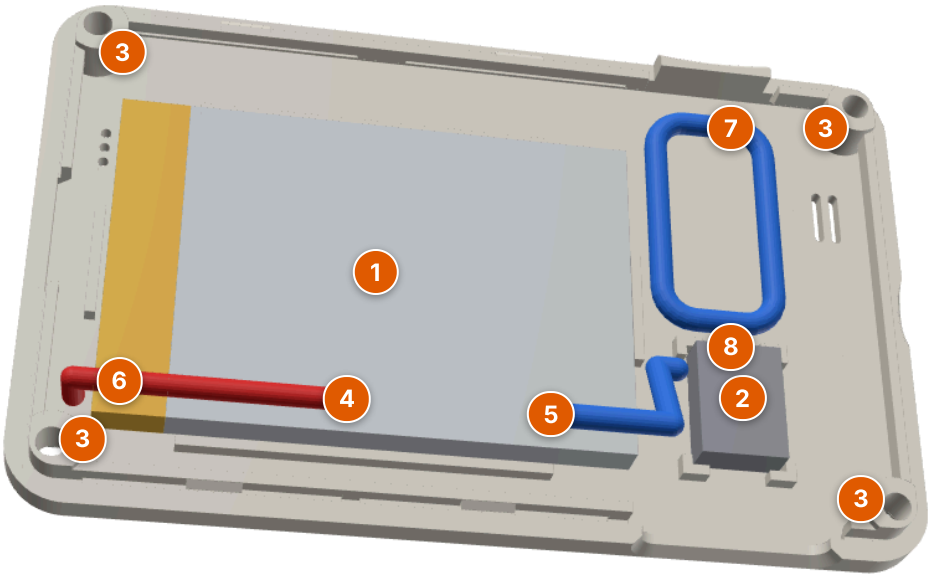



1. Dial notch (top edge).
2. PWR / BOOT opening (bottom edge).
3. USB-C opening.
4. The board, display side down.
5. microSD slot.

## 2 Battery, speaker and wires in the back cover

Lay the back cover inside up with the dial tab away from you, as in the picture, and lay the lid, with the board in it, just in front of it, **dial notch toward you**, so the two bottom edges touch. **Battery:** tape it into its bay with the **taped end (with the wire) toward the USB-C end**; if the wire comes out of the top corner, turn the cell over. **Speaker:** into its pocket **membrane side down** (white membrane in a black ring, facing the slots), **white peg toward the top edge**, so its cable faces the battery. Stick it with a frame of thin tape round the membrane, leaving the membrane uncovered. Plug the battery into **BAT** and the speaker into **SPK** (on the board's back, along its bottom edge) and run both wires along the bottom edge (about 34 mm and 27 mm). Tuck the spare battery wire beside the cell's taped end. Coil the spare speaker cable in one loop above the speaker and tape it down, away from the right-hand end where the antenna is.

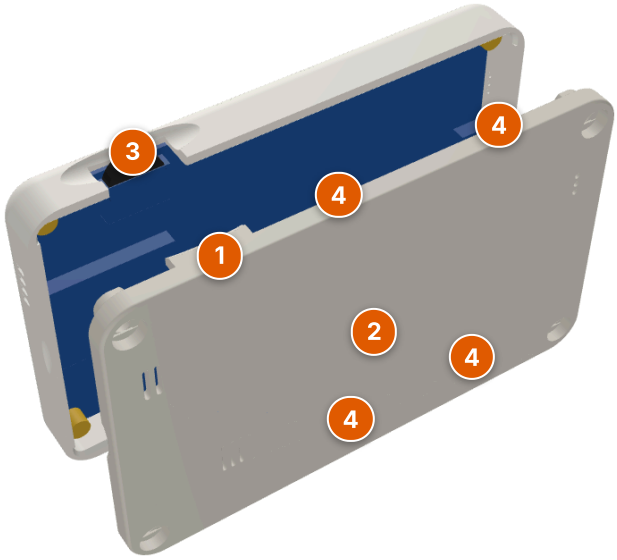
> **Check the connector:** BAT, RTC and SPK sit next to each other. The battery goes into **BAT** only. Keep every wire off the 4 corner posts (marked 3 in the picture below) and away from the long edges, where the clips are, so nothing gets pinched when the case closes.




1. Battery in its bay (wire end toward the USB-C end).
2. Speaker, membrane side down over the slots.
3. The 4 corner posts: keep the wires off them.
4. Battery plug → BAT connector.
5. Speaker plug → SPK connector.
6. The battery wire leaves the bay here.
7. Spare speaker cable: one loop, taped down.
8. The speaker's white peg points to the top edge.

## 3 Clip the back cover onto the lid

With the wires plugged in, **turn the back cover over onto the lid like the page of a book along the bottom edge**, where the wires run, so they aren't pulled. Before you lower it, **push both wires inside the lid's rim**, so that neither lies across the rim or the cover's bottom wall. Hold it over the lid with its **dial tab** above the slot over the dial, then lower it **straight down**. Press firmly along both long edges until **all 4 clips click** and the seam is closed all round. Then check that no wire shows in the seam or in the PWR/BOOT opening; if one does, open the lid again and tuck it in.

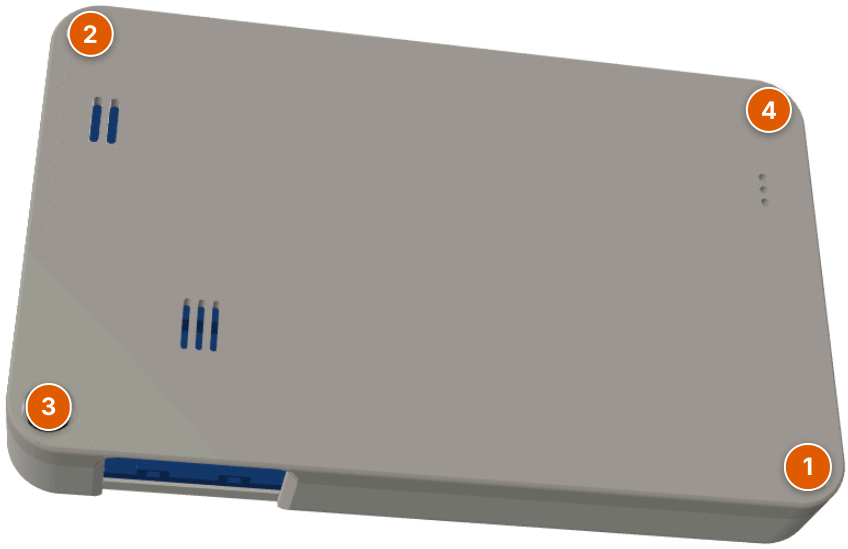



1. Dial tab: it goes into the slot above the dial.
2. Back cover: lower it straight down.
3. The dial in its notch.
4. The 4 clips, two on each long edge.

## 4 Drive in the 4 corner screws

The back now faces you. Drive one screw into each corner hole, into the board's brass standoff. Start each one by hand: an M2.5 turns in easily. **If it won't start without force, your standoffs are M2:** use the M2 × 8 mm screws, each with a washer under its head. If a screw just spins without catching, press it in gently as you turn: with 3 mm standoffs the board lies 1.5 mm from the posts until the screws pull it up. Tighten them in the order of the numbers in the picture (a cross pattern) with a small screwdriver turned by your fingertips, until each head sits snug in its recess and the board doesn't move.

> **Don't over-tighten**: the brass thread won't strip, but too much force can crack or squash the PLA posts. Use only 8 mm screws: shorter ones catch too little of the thread, longer ones can reach the board.




1. Corner screw 1.
2. Corner screw 2.
3. Corner screw 3.
4. Corner screw 4.

## Opening and reassembling




1 The opening notch in the end opposite the USB-C port, on the seam.

- To take the lid off, put a fingernail or a guitar pick into the **notch in the end opposite the USB-C port** (picture above), lever the lid away from the back cover, and work along both long edges until all 4 clips let go.
- **The board stays screwed to the back cover**, with the battery and the speaker. To take the board out, undo the 4 screws.
- To close it again, press the lid back on until the 4 clips click. The clips can be opened and closed many times.

## Tips

- The PWR and BOOT buttons stay fully uncovered in the bottom edge; press them straight on or slightly tilted.
- The speaker plays through the slots in the back. Lying on its back on a table the case sounds muffled: stand it on its bottom edge (not its top edge, where the dial is), or lay it face down.
- To eject the microSD card, press it in with a fingernail or a pen tip: it sits about 4 mm inside the edge.
- Your standoffs are 3–4 mm tall. With 3 mm ones the board sits 1 mm further back and the display 1 mm deeper behind the window; that's expected, and everything still fits.
- The standoff positions were read from your back photo and Waveshare's photos, which differ by up to 0.7 mm; the corner holes allow 0.4 mm, so the board can always sit with no screw more than 0.3 mm off centre. The battery and speaker sizes were read off the ruler in your photos (about ±0.3 mm). A test fit of the back cover is the real check.

Files: `stl/front_shell_battery.stl` (the front lid) and `stl/back_cover_battery.stl`. Details and every assumption are in README.md; all checks are in DEBUG_LOG.md.